

ECON 0100 | Homework B4 | Efficiency

Due: Sunday, Sep. 27

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d \quad (1)$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q \quad (2)$$

Prices are in galleons and quantity is in bottles.

Q1 | Total Surplus

What is the market equilibrium price, quantity, and total surplus? Use a graph and algebra to analyze this market.

P*: _____

Q*: _____

TS: _____

Q2 | Price Ceiling

The Ministry of Magic decided to institute a price ceiling of 50 galleons to allow many more to afford the treat. Use a graph of the demand curve and the supply curve to evaluate the welfare effects this policy had on the market.

Quantity Supplied: _____

Quantity Demanded: _____

Excess, Market Clearing, or Shortage: _____

CS: _____

Δ CS: _____

PS: _____

Δ PS: _____

DWL: _____

Q3 | Price Floor

The Ministry of Magic decided that the market price wasn't sufficiently high enough for sellers, and decided to institute a price floor to raise the price. Use a graph of the demand curve and the supply curve to evaluate the welfare effects of a price floor of 70.

Quantity Supplied: _____

Quantity Demanded: _____

Excess, Market Clearing, or Shortage: _____

CS: _____

Δ CS: _____

PS: _____

Δ PS: _____

DWL: _____